



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 04.08.2026

In addition to Part I (General Handout for all courses appended to the Time Table), this portion gives further specific details regarding the course.

Course No. : MSE G521
Course Title : WORLD CLASS MANUFACTURING
Instructor In-charge : ABHIJEET K. DIGALWAR

1. Course Description

The world-class manufacturing challenge, Developing world-class manufacturing strategy, Lean, Green, and Agile manufacturing strategy, Total quality management, Total employee involvement, Total productive maintenance, Decision-making methods, World-class information systems, Knowledge management, Managing the change, Monitoring the world-class performance.

2. Scope and Objectives

- To promote a good foundation in World-class Manufacturing,
- To explain the various concepts and philosophies in more detail for design of strategies for World-class Manufacturing,
- To develop familiarity with different decision making tools for evaluation and selection of best practices for world class manufacturing.
- To develop skills for implementation of World-class Manufacturing concepts and philosophies

3. Prescribed Text Book

- T1. Todd, Jim "World-class Manufacturing", McGraw Hill, London, 1995
- T2. Nicholas, John M., "Competitive Manufacturing Management", Tata McGraw Hill Education Pvt. Ltd. New Delhi, 2012.
- T3. Mohanty, R. P. and Deshmukh, S.G., "Advanced Operations Management". Pearson Education, 2003.
- T4. Besterfield D. H., et al., "Total Quality Management", Pearson Education, 1999.

4. Reference Books

- R1. Voss C.A., "Manufacturing Strategy: Process and Content", Chapman & Hall, London, 1992.
- R2. Research papers published in WCM domain (will be shared with students)
- R3. Nakajima, S., "Introduction to Total Productive Maintenance", Productivity Press, 1988.
- R4. Pascal, D., "Lean production simplified", 2nd Edition, Productivity Press, 2007
- R5. Digalwar, A.K., Tusnial, Anirudh (2020), "A systematic review of the integration of best practices for world class manufacturing", Int. J. of Business Excellence, Vol. 23 No. 3, pp. 353-388.



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- R6. Digalwar, A. K. and Date, Prasanna (2016), "Development of Fuzzy PROMETHEE Algorithm for evaluation of Indian World Class Manufacturing Organizations", International Journal of Services and Operations Management, Vol. 24, No.3, pp.308-330.
- R7. Digalwar, A. K., Jindal Anil. and Sangwan, K.S. (2015), "Modeling the Performance Measures of World Class Manufacturing using Interpreting Structural Modeling", Journal of Modelling in Management, Vol. 10, No.1, pp. 4-22.
- R8. Digalwar, A. K. and Nayagam, Padma (2014), "Implementation of Total Productive Maintenance in Manufacturing Industries: A Literature Based Metadata Analysis", The IUP Journal of Operations Management", Vol. 13, No.1, pp. 39-53.
- R9. Digalwar A. K. and Sangwan K. S. (2011), "An overview of existing performance measurement frameworks in the context of world class manufacturing performance measurements", International Journal of Services and Operations Management, Vol. 9, No. 1, pp.60-82.
- R10. Digalwar, A.K. and Sangwan, K.S. (2008), "Evaluation of world class manufacturing systems: a case of Indian automotive industries", International Journal of Services and Operations Management, Vol. 4, No.6, pp. 687-708.
- R11. Digalwar, A. K. Sangwan, K.S. (2007) "Development and Validation of Performance Measures For World Class Manufacturing Practices in India", Journal of Advanced Manufacturing Systems, Vol. 6, No.1, pp. 21-38.
- R12. Digalwar, A. K and Metri Bhimaraya A, (2005), "Performance Measurement Framework for World-Class Manufacturing", International Journal of Applied Management and Technology. Vol.3, No.2, pp.83 -101.
- R13. Digalwar A.K., Tagalpallewar A. R. and Sunnapwar V. K.(2013), "Green manufacturing performance measures: An empirical investigation from Indian manufacturing industry" Measuring Business Excellence Vol. 17, No. 4, pp. 59-75.
- R14. Digalwar, A. K., Haridas, P. and Joseph, Nelson I. (2014), "Development and validation of quality management constructs for software industries: an empirical investigation from India", International Journal of Services and Operations Management, Vol. 17, No.4, pp.453-478.
- R15. Digalwar, A.K., Mundra Nidhi, Tagalpallewar, A. R., Sunnapwar, V. K. (2017), "Roadmap for the implementation of green manufacturing practices in Indian manufacturing industries: An ISM approach, *Benchmarking an International Journal*, Vol. 24, Issue 5, pp.1386-1399.
- R16. Bhakar, Vikrant, Digalwar, A. K., Sangwan, K.S. (2018), "Sustainability Assessment Framework for Manufacturing Sector–A Conceptual Model", Procedia CIRP, Vol. 69, Issue 1, pp. 248-253.
- R17. Digalwar, A.K. (2018), "Application of Fuzzy TOPSIS for evaluating machining techniques using sustainability metrics", IOP Conference Series: Materials Science and Engineering, Vol 346, issue 1, pp. 1-8.
- R18. Gupta Mahesh, Digalwar A.K., Gupta Ajay and Goyal Animesh (2022), "Integrating Theory of Constraints, Lean and Six Sigma: A Framework Development and its Application", Production Planning and Control (DOI: 10.1080/09537287.2022.2071351.)

5. Course Plan

Module Number	Lecture session/Tutorial Session.	Reference	Learning Outcome
1. Introduction to World Class Manufacturing (WCM).	L1.1.Introduction to WCM, Course Objectives and outcomes		To understand the importance of world class manufacturing and how the evolution of manufacturing taken place
	L 1.2. Evaluation of Manufacturing from Craft Production to Sustainable Manufacturing	T1, T2 and R2	



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Module Number	Lecture session/Tutorial Session.	Reference	Learning Outcome
	L1.3. Emerging Trends in Manufacturing	T1, T2, T3 and R2	
2. Introduction to basic WCM tools and techniques	L2.1 Tools and Techniques (Planning and Design)	T3 and R2	To understand the various tools and techniques available for WCM practices
	L2.2.Tools and Techniques (Manufacturing and Distribution)	T2, T3 and R2	
	L2.3 Integration of best practices	T3 , R2, R5 and R18	
3. Manufacturing Strategy.	L3.1 Operations strategy, strategic approaches to manufacturing,	R1	To understand various manufacturing strategies and to learn how to formulate and implement the strategy
	L3.2 Application of benchmarking processes	T4 and R2	
	L3.3 Process of manufacturing strategy formulation and implementation	R1	
	L3.4 Case studies and applications	T3, R2 and R10	
4. Decision making tools ,techniques and methodologies	L4.1 Introduction to decision making tools, MCDM, MADM and MODM	R2	Understanding the various decision making tools, techniques and methodologies and their applications
	L4.2. AHP, TOPSIS, ISM, etc.	R2	
	L4.3 Fuzzy Set of Theory, Integrated and Hybrid Approaches	R2	
	L4.5 Problem solving	R2, R6, R7 and R10	
5. Total Quality Management	L5.1. Introduction, Quality evolution, tools and techniques,	R2 and T4	To understand the importance and applications of TQM for WCM practices.
	L5.2. principles and processes of TQM formulation and implementation	R2 and T4	
	L 5.3 TQM Frameworks	R2 and T4	
	T5.4 Case studies	R2 and R15	
	L6.1. Introduction to lean and Agile Manufacturing	T2 and R4	To understand the importance and



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Module Number	Lecture session/Tutorial Session.	Reference	Learning Outcome
6. Lean and Agile Manufacturing	L6.2. Lean Principles, tools and techniques	T2	applications of LM for WCM practices.
	L 6.3 Lean manufacturing implementation frameworks	R2	
	L6.4 Benefits of Lean and Agile Manufacturing	R2	
	L6.5 Case studies	R2	
7. Total Productive Maintenance	L7.1. Introduction to various maintenance system, overview of TPM,	R2 and R3	To understand the importance and applications of TPM for WCM practices.
	L7.2 TPM Pillars, Tools and Techniques	R3	
	L 7.3 TPM implementation frameworks	R2	
	L 7.4 Case studies and meta data analysis	T3, R2 and R8	
8. Other aspects of WCM	L8.1 Overview of sustainability aspects	R2	To understand the advances and recent developments in the manufacturing organizations
	L8.2. Overview of Green Manufacturing and circular economy	R2, R15, R16 and R17	
	L8.3 Overview of sustainable supply chain	R2	
9. Corporate Social Responsibility	L 9.1 Role of CSR in WCM	R2	To understand CSR importance and need
10 Industry 4.0	L10.1 Industry 4.0 and readiness of Indian manufacturing industries for it	R2	To understand Industry 4.0 and readiness of Indian manufacturing industries
11. Standards, Certifications and Performance Measurement Systems	L 11.1 ISO standards	T4 and R2	To understand the various standards, awards, certifications and PMS and their importance for WCM
	L11.2 Various awards, recognitions, certifications, criteria for awards and certifications	T4 and R2	



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Module Number	Lecture session/Tutorial Session.	Reference	Learning Outcome
	L11.3 Overview of performance measurement systems	R2 and R10	
	L11.4 Comparison of various PMS for WCM	R2 and R10	
	L11.5 Development and validation of PM for WCM	R2, R11 and R12	

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 Min.	25%	05-10-26 AN2	CB
Comprehensive Examination	180 Min.	40%	03-12-26 AN	CB/OB
Case Discussions/ Group Projects/ Surprise Class Tests/ Assignments		35%	TBA	CB/OB

7. Chamber Consultation Hour: Monday, 10th hour chamber: 2227

8. Notices: All notices regarding the course will be displayed on the **Mechanical Engineering** notice board.

9. Make-up Policy: No make-up for class tests and non-appearance in class tests may lead to NC.

10. Note (if any):

Instructor-in-charge
Course No. MSE G521